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Faculty of statistics and informatics



Design of an electronic election system using blockchain technology

BACHELORS THESIS

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Prohlášení

Prohlašuji, že jsem bakalářskou práci Návrh elektronických voleb založených na technologii blockchain vypracoval samostatně za použití v práci uvedených pramenů a literatury.

V Praze dne 10. května 2021

.....

Vladislav Tkačík

Poděkování

Tímto bych chtěl poděkovat mému vedoucímu práce Ing. Tomáši Brucknerovi, Ph.D za odborné vedení a trpělivost při vedení této práce. Dále bych také rád poděkoval lidem z firmy Tatum, kteří byli ochotní mi věnovat svůj čas a poskytl mi mnoho užitečných rad, které mi pomohly při vypracovávání této práce.

Abstrakt

TKAČIK, V. Návrh elektronických voleb založených na technologii blockchain.
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Tato bakalářská práce pojednává o problematice elektronických voleb a návrhu vhodného elektronického volebního systému využívající technologii blockchain pro Českou republiku a její parlamentní volby. Obsahem práce bude rešerše problematiky elektronických voleb a dále část praktická, která se zabývá již samostatným návrhem volebního systému.

První částí a zároveň také prvním cílem je poskytnout informace a znalosti, které poslouží autorovi pro vypracování praktické části práce. Tyto znalosti autor nabude důkladnou rešerší problematiky elektronických voleb a prostudováním již dříve realizovaných volebních systémů využívající blockchain i jejich návrhů.

Druhým cílem je za pomoci nabytých znalostí vytvořit vhodný návrh volebního systému a jeho implementaci. Autor provede analýzu, díky které bude moci tohoto cíle dosáhnout.

Klíčová slova

Elektronické volby, volby, blockchain, blockchain volby, návrh blockchainového volebního systému, digitální volební systém

Abstract

This thesis deals with e-voting and designing an appropriate e-voting system based on blockchain technology for elections in the Czech Republic and its Parliament election. This thesis consists of a research of e-voting systems and a practical part, which consists of designing the voting system itself.

In the first part, the first goal is to provide information and knowledge upon which the practical part of this thesis is based on. The author will acquire this information by an in-depth analysis of electronic voting and by studying already realised e-voting systems as well as voting systems based on blockchain and also their design proposals.

The second goal is to create a suitable design of a voting system based on blockchain by using knowledge acquired in the first part.

Keywords

Electronic elections, elections, blockchain, blockchain elections, design of a blockchain election system, digital election system

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1 Introduction

Ancient and modern human civilizations have always tried to tame elements, that nature had to offer, as well as different materials and technologies, so that they could perform specific tasks more easily and more effectively. The evolution of human society brought us many technological innovations, which in their time mostly reflected the very essence of the population's needs and culture. Therefore, we can easily say that these technologies and human societies are becoming more and more inseparable from each other as time goes on. (Wardynski, 2019)

Right now, we are living in the 21st century, which is known for its technological advances that are being applied to aspects of our everyday lives, without which we most likely could not function as effectively as we are able to now. But there is one specific technology that has revolutionized the way we communicate on our daily basis. That technology is of course no other than the internet. The internet allows us to connect and communicate with people on a whole new level and also digitalize many aspects of our lives. One of these aspects is also money, which happened to be fully digitalised in 2009 when bitcoin was created. However, we will not talk about bitcoin but rather a technology called blockchain, on which bitcoin is made upon. Blockchain technology has great potential in foreseeable future and remains highly discussed topic among tech companies that try to find appropriate uses cases for this technology.

There is also one aspect of our lives which has not yet been fully affected by digitalisation. And that aspect is elections. Do current technologies offer us some ways to make electronic voting possible without undermining the credibility and reliability of election process, which is the main mechanism of representative democracies? And could blockchain be the solution to some of the problems the current election systems used around the world face? The author will try to elaborate on these questions throughout this thesis.

1.1 Thesis goal

The main goal of this thesis is to propose a new design of electronic voting system for Czech Republic Parliament election, which will use blockchain technology. The ideal outcome would be to find a way to vote using our devices at home, such as smartphones or laptops, as this might attract many lazy people that do not usually bother to express their opinion by voting. However, it may or may not be an appropriate idea at all.

The second goal is to provide comprehensive documentation of currently established voting systems in the world and their implementations. The author will use necessary information to propose an ideal voting system that would have better chances to defend election process from possible tampering that could endanger the democracy itself. Author will also cover the background of blockchain technology and its functionality.

1.2 Methodology

At the beginning, the author will focus on gathering all necessary information that will serve as a basis for better understanding of all established processes and practices behind

democratic elections in general. He will research different types of voting systems in order to determine which one would suit proposed design the best.

Then he will be looking more closely into current election system in the Czech Republic. He will also focus on possible ways how could this system be tampered with. If so, the author will analyse if there are any possible solutions to these flaws. The final proposition would then contain the solutions to these problems using combination of modern technologies used elsewhere in the world, as well as blockchain technology.

The author will also analyse and compare different election systems based on blockchain and their design propositions.

1.3 Anticipated outcomes

The outcomes of this thesis are based on its goals and methodology, which the author chose to approach. In the first part we can expect outcomes such as:

- Comprehensive description of democratic elections and all its necessary parts
- Comparison of different types of election systems established in the world
- Description of Czech election system
- Description of already implemented election system designs using blockchain technology

In the second part, the author then focuses on his own design proposal. Anticipated outcomes are:

- Design of an election system to be used for Czech Republic Parliament election
- Used technologies
- Implementation of proposed system
- Other possible use cases

2 Blockchain

These days the word blockchain may not be so foreign to people now as it might have been a few years back. However, if you have not come across the word blockchain yet, then you might have most likely heard words like Bitcoin or Ethereum. Those are currently the most popular cryptocurrencies among hundreds of others, which are based on blockchain technology. Many people might think, that blockchain and bitcoin are the same thing, because these two terms are being so closely related. Nevertheless, that is a misconception. These two terms are indeed different, and the author will shed some light on it in a short moment. In this thesis, terms like bitcoin, blockchain, cryptocurrency and similar terms related to this topic will occur, so defining these terms would be appropriate now, in order to correctly understand the problematics and not to cause any confusion.

There are plenty of definitions of blockchain, written by many companies and people, as each individual or company may view it from a slightly different angle. Gartner, which is the world's leading research and advisory company, has defined blockchain as:

„A blockchain is an expanding list of cryptographically signed, irrevocable transactional records shared by all participants in a network. Each record contains a time stamp and reference links to previous transactions. With this information, anyone with access rights can trace back a transactional event, at any point in its history, belonging to any participant. A blockchain is one architectural design of the broader concept of distributed ledgers.“ (Definition of Blockchain - Gartner Information Technology Glossary, no date)

One of the most typical uses of blockchain is in cryptocurrencies. The general definition of cryptocurrencies may look like:

“A cryptocurrency is a digital or virtual currency that is secured by cryptography, which makes it nearly impossible to counterfeit or double-spend. Many cryptocurrencies are decentralized networks based on blockchain technology—a distributed ledger enforced by a disparate network of computers. A defining feature of cryptocurrencies is that they are generally not issued by any central authority, rendering them theoretically immune to government interference or manipulation.”(Frankenfield, 2020)

Utilizing the blockchain technology led to the invention of the first decentralized cryptocurrency, bitcoin.

2.1 History

As early as 1991, scientists Stuart Haber and W. Scott Stornetta managed to describe the idea behind blockchain technology when they found out a way how to time-stamp digital documents by using computationally practical solution in a way that the documents could not be backdated or tampered with. The system was designed in a way that it used cryptographically secured chain of blocks where each time-stamped document was stored in a separate block. The next year they implemented new design, which allowed them to store multiple documents in a single block. Nevertheless, in that time it did not seem to be such a breakthrough that would solve any significant problems and after several years, the patent for this technology lapsed in 2004. (*History of Blockchain*, 2020)

In 2004, a system called RPoW, Reusable Proof of Work was introduced by cryptographic activist Hal Finney. It was a software which allowed us to reuse POW (proof of work) token. "A POW token is something that takes a relatively long time to compute but which can be checked quickly. Normally POW tokens can't be reused because that would allow them to be double-spent." (*RPOW - Reusable Proofs of Work*, no date) Meaning that we could use one token for multiple transactions which would mean duplicating the token. RPOW token on the other hand can be reused, but its reusability is limited and can be reused only sequentially. This means that the system allows the POW token to be used once, then the token must be exchanged for a new one, which then could be used once more, then again exchanged etc. This is managed by a server which keeps records of the tokens and their owner by assigning the owners a private key to the token. The owner of the token can then give it to someone else by signing the token to some public key via transfer order. The token is then registered by the server and is assigned to a corresponding private key. This approach solved the double-spending problem by keeping track of the ownership of the tokens on a trusted server and is fundamental to all digital cash. (*RPOW - Reusable Proofs of Work | Satoshi Nakamoto Institute*, no date)

In 2008 a new concept of digital cash was published, in which Satoshi Nakamoto, a person or a group of people working under this pseudonym, introduced whitepaper describing Bitcoin currency in detail. Bitcoin was then implemented in early 2009 when the first block of bitcoin was mined by Satoshi Nakamoto himself. (*History of Blockchain*, 2020)

Bitcoin was not the very first cryptocurrency created, but it solved several problems concerning digital currencies and that is why it is the most successful cryptocurrency today.

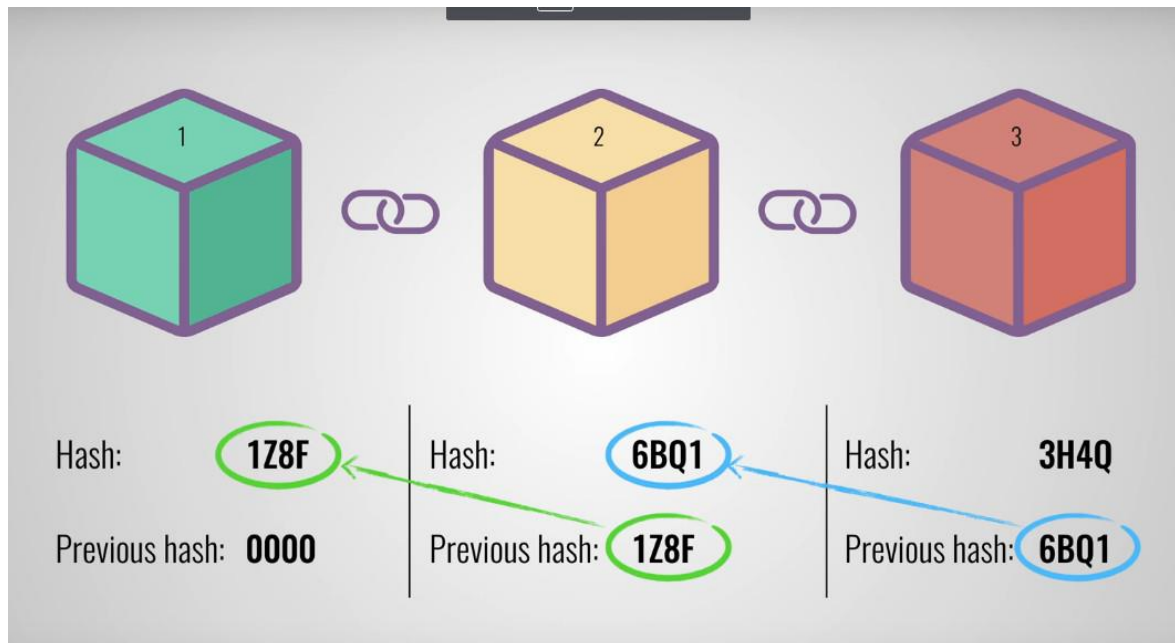
2.2 Functionality

As we have previously defined, blockchain is a type of database where we can store all sorts of data. Classical databases use different approach to store data and that is by using tables. These tables filled with data are usually stored and centralized on one computer or within one organization which gathered the data. On the other hand, blockchain stores data into blocks in decentralized way. These data are stored on distributed network of computers which maintain the shared database thus there is no central authority which would have to protect and store all the data.

Each block can store only certain amount of data. After this data is gathered, they form a block which is encrypted and given a hash value. No matter how much data is stored in the block, be it a "Hello World" string, a string of a random characters like ",-/H3l0+." or a whole book, the encrypting algorithm would always compress this data into fixed length hash consisting of 64 characters. This hash value serves as a unique identifier for each block. Once everything inside a block is encrypted, none of the data can be reverse calculated from the hash value. However, we can always run the data in a block through an algorithm to check if the hash matches and data were not compromised somehow. Changing only one letter or a random character in some data stored in a block would lead to making a completely new hash. (Coding Tech, 2018)

Every time a new block is created it is placed at the end of the chain of all previously created blocks. This new block is securely linked with the previous one by containing the hash value of previous block and is also timestamped with a time when it was created. Each block then contains its own hash, hash of previous block, timestamp and data.

If we were to tamper with one of the blocks and change some data inside hash value of the block would change. It would then not correlate with the following block and therefore, it would alter all the following blocks of the chain. We would have to recalculate and change the hashes of all of those blocks to make the whole chain valid again. This would not be such a problem for most modern computers which can produce huge amount of computing power, but blockchain has two additional security features, which makes it nearly impossible to break. Those features are decentralization and proof-of-work (PoW).



Picture 1 - Blockchain scheme with hashes (Kiki Ahmadi, 2018)

Creation of new blocks is slowed by PoW mechanism which needs to be done to create a new block and add it to chain. This takes around 10 minutes on bitcoin blockchain and around 20 seconds on Ethereum blockchain. Changing only hashes of the blocks would not be enough and we would also need to compute all the PoW for each tampered block which would need enormous computing power. Even if we would somehow manage to recalculate all the tampered blocks which is unlikely, then these changes would have to be accepted by all of the nodes on the network. When new block is created, it is sent to all of the nodes to validate if it suits their version of blockchain. If more than 50% of all nodes agree, that this is indeed valid block, only after then it is added to the blockchain. (Simply Explained, 2017)

The only possible way would be to tamper with a block, recalculate all the hashes and PoW for each block and also do this on more than 50% of all the nodes on the network so an attacker would create a “false consensus” with which he would make his version of the blockchain valid. Tampering with blockchain data stored inside is therefore nearly impossible. (Simply Explained, 2017)

The last thing worth mentioning is that blockchain uses Public-Key Cryptography. Each user has its own public and private key pair which is generated alongside with creation of digital wallet when joining blockchain network. Public key serves as an address where other people from the network can direct their transactions and send you some tokens which represent money in a form of some amount of bitcoin when on blockchain network. Public key can therefore be seen by anyone without disrupting anything from your wallet which serves as a safe where all of your tokens are stored. The only way how to get in is to

own the correct private key. So, unless you lose the private key, only you can successfully get into the digital wallet.

2.3 Voting systems using blockchain

There are many projects of different sizes which have already tried to implement blockchain into a voting system. The most well-known projects are Voatz and Agora. Helios, which does not use blockchain directly but offers end-to-end verifiability of the whole election process using cryptographic protocols, is also highly popular. (Villalobos, Altamirano and Chandra, no date)

Helios

Helios is a web-based, open-audit online voting system. This system is currently accessible on their website and is allowing anyone to use it in order to create and run an election. It is designed especially for smaller elections such as local clubs, student governments and other smaller groups where it is necessary to preserve secrecy and trustworthiness of the election, but coercion would not be a huge problem. (Adida, no date) Therefore this system is not particularly supposed to be used for nationwide elections. Creators of this system also do not think it is suitable for any state elections but not that they would consider Helios as unsecure, but rather they do not believe people's devices at home to be secure enough in case of significant attacks. (*Helios Voting*, no date)

Voting works this way (Villalobos, Altamirano and Chandra, no date):

- a) Election is created by an administrator who enters the names of all candidates as well as time when election begins and ends. Then he creates a list of emails (eligible voters) to which credentials needed for voting (username, password) are sent. Administrator has no access to these credentials.
- b) Once voter receives the email, he uses the credentials to log into interface and casts his vote. Voter has then option to check if the ballot is being cast correctly.
- c) At the end of the election each voter is informed about their vote being tallied.

Voatz

Voatz is a mobile elections platform. It is run on smartphones and requires the latest, most up to date operational system software to be able to use this platform. This platform's security is based on security of smartphones and the tamper-proof design of blockchain. People who have older phones, which are not supported, are not eligible to use it. The entry barrier is therefore quite significant, but it is the price for having relatively more secure election. (Villalobos, Altamirano and Chandra, no date)

Voting works this way (Villalobos, Altamirano and Chandra, no date):

- Voatz is given access to a database with information about all of the voters by election officials.
- Voters need to authenticate first by scanning their ID, sending their live selfie and fingerprint as well. App then checks the validity of fingerprint by matching it to the device as well as verifying validity of provided selfie and ID.

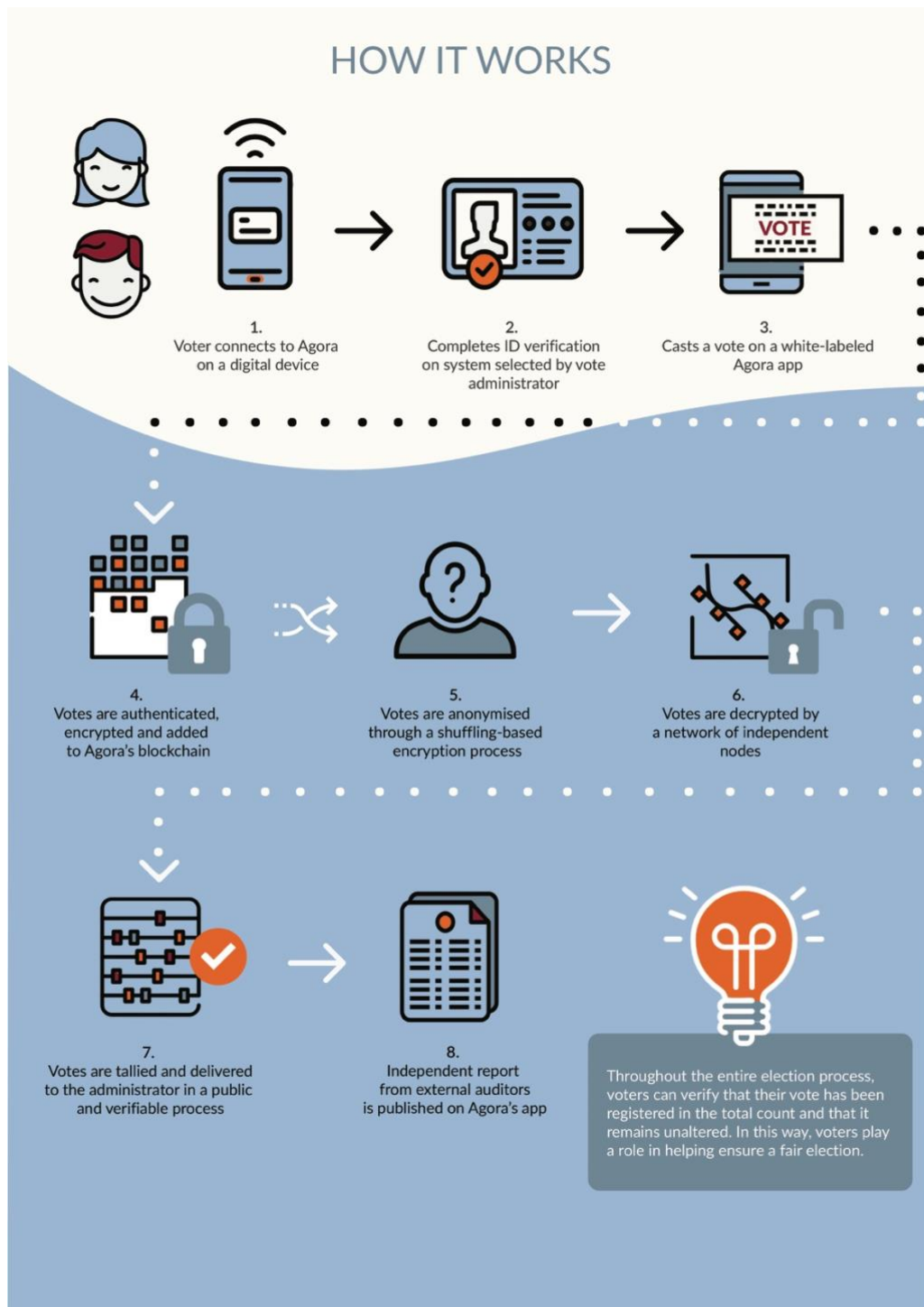
- After successful verification users can vote. Cast vote is then represented as a transaction on a blockchain. This transaction has to be verified by a specific number of servers.
- Ballots are stored on blockchain until Election day when they are physically printed and counted with the rest of the ballots. (Visram, Visram and Visram, 2020)

Voatz has already been piloted in a number of states in US such as West Virginia, Colorado and Oregon. Utah had the honour to hold first live demonstration of the way how Voatz ballots can be audited. In 2019 Utah allowed their military voters to use Voatz instead of using email which is considered to be way more unsecure. This option was also allowed for disabled people. (*Utah County Makes History With Presidential Blockchain Vote*, 2020)

Agora

Agora is a voting ecosystem using blockchain technology which allows people to vote online with their devices in an easy and certain way. Voting is end-to-end verifiable and offers complete, auditable, undisputable, open and transparent record of election on a custom blockchain. Agora also provides different solutions other than voting app to fit different needs such as voting protocol and supply chain tracking system. (*What is Agora Page*, no date) In order to be able to fully implement the Agora voting system, a country needs to have a digital identity system or biometric database for their voters established. ('Agora - Digital National Elections with Blockchain', 2018)

Agora acted as an international observer during Sierra Leone presidential election in March 2018. Agora was given a status of an observer by the National Electoral Commission (NEC). The goal was to demonstrate capabilities of their system to open the doors for further cooperation. Agora was given a permission to cover the Western Districts with 280 polls. Official results of the election were conducted only by NEC itself. In the end, the results from Agora and NEC were very similar, but agora had them published 5 days faster than NEC. (Team, 2018) Voting flow using Agora voting app can be seen in Picture 2 down below.



Picture 2: Agora voting ('Agora - Digital National Elections with Blockchain', 2018)

3 Elections

An election is a process by which people select a person who strives to represent some public office. These people are chosen to make important decisions in people's interests. Therefore, they represent the power of people to decide on specific matters. It is also a way by which people can accept or reject some political propositions and this is done by voting. (Sara Pennicino, 2017)(Paul David Webb, 2020). Elections are essential in each modern democratic country and it is the fundamental on which all of these countries are built on. Elections should contain both the election form as well as the election substance to be deemed democratic. If one of these requirements is missing, then we would lean towards different political systems. In case of a missing election form we would lean towards constitutionalism of illiberal regimes. On the other hand, if an election substance is missing, that would mean that either we are in the presence of attempt to transform democracy into an authoritarian regime or liberal democracy would be degraded. (Sara Pennicino, 2017)

There are two forms of election in democracy which is either direct or indirect. That depends on whether voters elect people who represent them and decide on their behalf as we have partly covered in the first paragraph, or whether people are directly a part of the political decision-making process thanks to which they could decide on specific propositions themselves. (Sara Pennicino, 2017) This is done by instruments such as referenda and initiatives. (Theo Schiller, 2020) However, fully direct democracy exists only in the Swiss cantons of Appenzell Innerrhoden in its pure form. In the rest of the world we can see semi-direct democracies (they are representative democracies with elements of direct democracy). (Vincent Golay and Mix & Remix, 2008) The most popular country which managed to preserve this type of political system is Switzerland. Their system has its roots as far as the middle of the 19th century. First changes on national level happened in 1848 when they introduced mandatory referenda. Every few years they kept adding more instruments of direct democracy which remained up until now. Some of the instruments of direct democracy can be seen in Easter European countries such as Latvia and Lithuania. (Theo Schiller, 2020)

However, direct democracy elections and voting will not be covered in this thesis by author, but some of the outcomes might be applicable to this type of election system as well. Instead, the author will focus on modern indirect democratic elections which are voluntary, free, fair, and equal. This form of election is currently used in Czech Republic and will be the main topic author will focus on.

3.1 Election system

Election is a complex process because they not only cover the act of voting but also the very process through which voters can actually cast their votes. For results of election system to be democratic, the method used to elect our representatives must be democratic as well. There is no general law which would describe elections as such because various elections in countries exist usually divided by their size into national, regional, provincial. Over the years however, there were some standards developed which are "considered to be inherent to the idea of election. At a domestic level, these standards are codified in constitutional, statutory, and regulatory provisions; however, in recent years, due to the

growing engagement of the international community in democratization processes, elections have become one of the focuses of international law.”(Sara Pennicino, 2017)

Election system is a method and set of rules which describe a way how votes are counted. By using these established methods, the final results of the elections are being determined. It exists several different systems. The most commonly systems used are plurality, majoritarian and proportional.

3.2 International Standards of elections

Defined by the Universal Declaration of Human Rights from 1948, each individual has the right to express his will which is to be the basis of authority of government. This will is to be expressed by holding periodical and genuine elections, where secrecy of votes has to be ensured as well as universal and equal suffrage. (*Universal Declaration of Human Rights*, 2015)

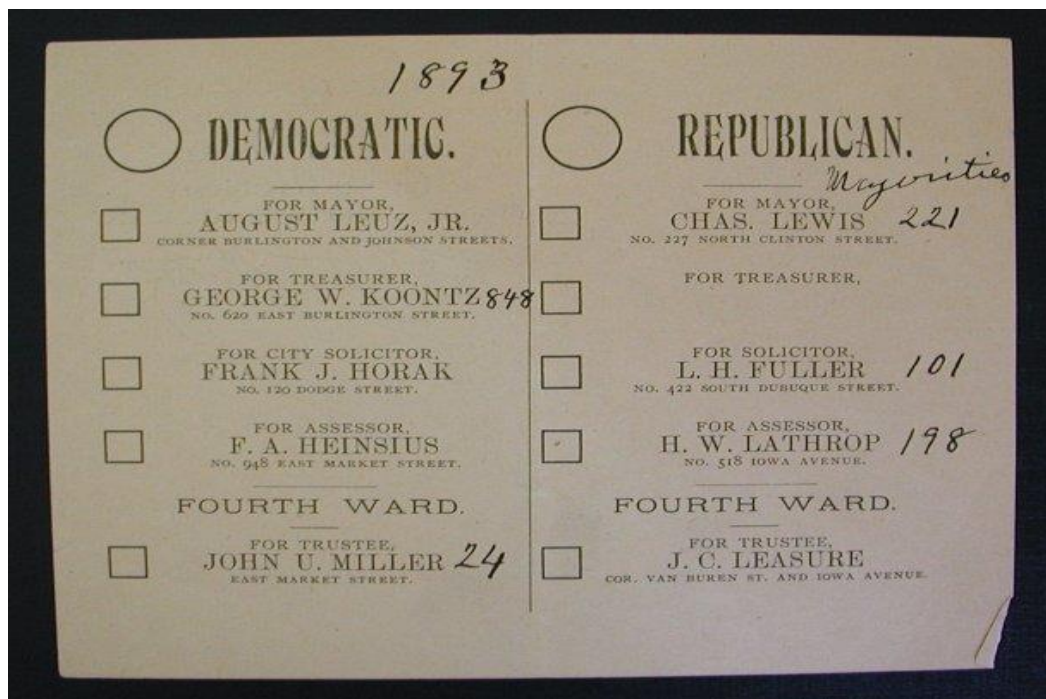
In 1990, the Council of Europe established European Commission for Democracy through Law, better known as the Venice Commission. “The role of the Venice Commission is to provide legal advice to its member states and, in particular, to help states wishing to bring their legal and institutional structures into line with European standards and international experience in the fields of democracy, human rights and the rule of law.” (*Venice Commission :: Council of Europe*, no date) In 2002, the Venice Commission released a document “Code of Good Practice in Electoral Matters” which defines general principals, that has to be followed in order to maintain Europe’s electoral heritage. These principals are: (‘CODE OF GOOD PRACTICE IN ELECTORAL MATTERS .pdf’, no date)

- Universal suffrage – each individual human being has the right to vote as well as the right to stand for elections. This individual, however, must meet certain conditions regarding age, nationality, residence and deprivation of the right to vote and to be elected.
- Equal suffrage – to preserve this principal, each voter has to have one vote. If electoral system allows to have multiple votes, each voter must have equal number of votes.
- Free suffrage – this principal ensures that voter is not forced to form an opinion. Opinion has to be made willingly and freely. It also allows voters to make actions if they think the elections were tampered.
- Secret suffrage – secret voting is not only right of all voters it is also mandatory part of the process. Any ballot that does not meet this criterion of secrecy is considered invalid. Voting must be also individual. Any form of controlled voting such as family voting must be forbidden.
- Direct suffrage – this suffrage makes sure that some authorities are elected directly.
- Frequency of elections – principal of frequency ensures that elections are held periodically, and that legislative assembly’s term of office must not exceed five years.

Each democratic country should therefore have appropriate election system design, that would honour all of these principals. Different approaches to election system designs and voting methods used in the world will be analysed and described in chapter 4.

4 Voting technologies

The election process during which the people vote for different matters has been here since ancient Athens, and during this time, people tried to invent different approaches regarding voting methods. The very first use of paper ballots appeared in Rome in 139 BCE. (*Douglas W. Jones Illustrated Voting Machine History*, 2003) However, in 1856, the Australian state of Victoria became the first country to unify the form of the ballots used for voting, later called Australian Secret Ballot. These ballots were printed at government's expense. (ProCon.org, 2017) This form of voting was adopted by other countries as well in the following decades.



Picture 3: Australian Ballot adopted in 1893 by Iowa City (*Douglas W. Jones Illustrated Voting Machine History*, 2003)

After the paper ballot method, people in 1889 invented a punch card system which laid the foundation for system using this very same technique in the 1960's. In the same year a new design of mechanical lever machine was patented. This machine had its positives as it prevented overvotes to be cast. It also speeded up vote counting process and reduced the chance of dishonest vote counting which could be caused by human factor. Then Optical Scans were used for the first time in 1962 followed by Direct Recording Electronic (DRE) Voting Machine patented in 1974. (ProCon.org, 2017) These last two voting methods are in some modernized ways used nowadays as well, mostly in the U.S.



Picture 4: Mechanical lever voting machine from 1899

Some countries pushed voting possibilities even further, like Estonia, which has already established the possibility to vote via internet, using any computer with internet connection anywhere in the world. Estonia was the first country in the world to hold nation-wide elections using internet-based voting method. (*i-Voting*, no date)

However, the most popular voting method still without a doubt remains manual marked paper ballot which is used in 209 of 227 countries for which the ACE Electoral Knowledge network was able to gather data. Other voting methods than paper ballot, which are mentioned in the previous paragraphs, are not as common and are implemented in only 10% of countries and territories for which the data is available. These countries usually have a mix of different types of voting methods. (Shannon Schumacher and Aidan Connaughton, 2020) The author will try to elaborate and provide more detailed description of all of these methods in the following sections.

4.1 E-Voting

Electronic voting, in short, e-voting, is a voting method which uses information technologies that help voters to go through election process. It is necessary to define the steps of this process because these systems might or might not be designed to cover all of these steps. Election process consists of four basic steps: (René Peralta, 2016)

- Ballot composition - this is the step where voters choose between the candidates and make their choice
- Ballot casting – this step allows the voters to cast their ballot using this system

- Ballot recording – in this step the system records and saves the submitted ballots
- Tabulation – this step ensures that all the submitted ballots by the system are counted

Sometimes such system also covers the step of authentication of the voter, but this varies based on the machine that is used. By following e-voting definition, it is possible to use devices like smartphones, laptops, desktop computers as well as DRE machines and optical scans.

There are mainly two different approaches to these system designs. Designs, that do not need access to the internet (e-voting) and designs that use the internet (I-voting). (René Peralta, 2016) All of e-voting systems are situated in polling stations where they are under supervision of the election commission. E-voting therefore needs voters to come to the polling place and vote in here the same way as they would vote with paper ballots. On the other hand, i-voting takes place in unsupervised areas because voters can use their smartphones or other devices that are connected to the internet to cast their votes. This approach could open up some other potential issues, which would be unlikely to happen at the polling station.

4.1.1 DRE Machines

Functionality

DRE Machine usually consists of a touch screen which is connected to a computer. This touch screen allows the voter to choose their candidate and cast a vote. It also opens another possibility to help the voter throughout the election process, as there are options available to display larger fonts or make the screen and text more contrasty to people with limited vision. It alerts the voter to not miss any steps and prevents him from overvoting. After the vote is cast and submitted, it is recorded and stored in the machine's memory. (René Peralta, 2016)



Picture 5: The Diebold AccuVote-TS voting machine (*Security Analysis of the Diebold AccuVote-TS Voting Machine*, no date)

These machines are sometimes called “Black boxes” because most of the things happening inside the machine are not observable. Ballot recording cannot be observed, and we have to fully rely on the machine and its software which performs this step of storing the ballot and voter’s intention correctly. With electronic devices, there is always the potential threat of some malicious software. This software could sit in the machine from as early as development stage of this machine’s software undetected and could get into the final machines used in elections. However, software used in these machines is usually kept away from public as companies tend to say, that the way these machines work is a trade secret. (René Peralta, 2016) (J. Alex Halderman, no date) The other reason not to share this information is from security standpoint, to keep the machines safe, they say. Not only the way how the votes are submitted is kept away from public, which results in a lack of transparency, but also this approach to security: “security by obscurity is widely discredited practise in software development.” (J. Alex Halderman, no date) Security through obscurity is bad because as soon as someone learns how to break into the system, they compromise it. (Daniel Miessler, 2019) “The only way to ensure that something is going to be robustly secure is to assume, that it is going to be available to the attacker.” (J. Alex Halderman, no date)

Auditability

Since there is no way how to verify votes cast in these machines, many people and organizations argue that some form of physical paper trail record has to be made once voters cast their votes. As response to this, manufacturers added small printer that would print out a cast vote on small piece of thermal paper. (René Peralta, 2016) This method is called Voter verified paper audit trail (VVPAT). However, with this, problems came regarding ways how to manage these paper ballots in order to protect ballot secrecy as well as storing them. Storing these ballots was a real challenge because of low quality papers used, which would fade away when exposed to sun for some time or just fade away with time. They were also hard to read when creased. (J. Alex Halderman, no date)

In India, they managed to make VVPATs work. They use dedicated VVPAT printer machine which is connected to electronic voting machine (EVM). When vote is cast VVPAT prints the vote and only voter is able to see the printed vote. Vote is shown inside the machine and cannot be interacted with and is displayed to a voter for 7 seconds to verify, if the vote was cast as demanded. After 7 seconds printed vote is cut and falls into the box among other ballots followed by a beep sound, meaning that ballot was counted. (*What is VVPAT? How does VVPAT work?*, 2020) Secrecy is therefore preserved because ballots are scrambled inside.

Security

Most of these DRE machines are not connected to the internet these days. This is great, because it cannot be attacked remotely on a massive scale. If these machines were connected to the internet, an attacker could tamper with many of these machines and maybe penetrate even further into the network without notice. There were some cases of these machines being connected to the internet while they were not in use and stored in a warehouse.

This additional layer of protection and auditability made by VVPAT is great because it makes tampering much harder. The attacker would have to tamper with both digital and physical ballots to match each other in order to successfully manipulate the election

results. This probably could not be done by a single person from comfort of his home unlike tampering with digital ballots only.

4.1.2 Optical scanning machines

Three main types of optical scanning technologies exist. Optical Mark Reading (OMR), Optical Character Recognition (OCR) and ICR which stands for Intelligent Character Recognition. The most popular technology among optical scanning voting systems is OMR. In US sometimes called as “Marksense” systems. (*Electronic/Mechanical Voting Systems* —, no date)

This machine usually consists of a touch screen, scan, and a locked storage box where ballots are stored once cast.



Picture 6: DS200 Optical Scan Machine (Voting Equipment Information | Elections, no date)

Functionality

The voter is given a sheet of paper with candidates where next to each candidate is usually an unfilled symbol, such as triangle, circle or any other geometric shape. The voter then chooses the candidates and fill incomplete symbols with pen or pencil, that correspond to the chosen candidates. (*Electronic/Mechanical Voting Systems* —, no date) This ballot is then ready to be cast into the optical scan. If everything is marked properly, the vote is accepted by the machine and is ready for the next vote to be scanned. If not, the machine shows an error message with instructions what needs to be done. The voter can then follow the instructions and make changes to the ballot or continue with his original candidate selection, which will not be counted. If he chooses to make changes to the ballot, it is then returned back from the machine, or he can take a new ballot, if he cannot make wanted changes to the original one. (*DS200 Digital Scanner Voting Machine - YouTube*, no date) In some cases, the symbols are not filled properly, and optical scan therefore may

or may not successfully scan a ballot. Data obtained from the ballot is digitalized and stored within the machine in local memory.

To avoid optical scans not reading ballots properly a second device is sometimes used that is called ballot-marking device (BMD). It is used instead of letting voter to mark a ballot by hand using a pen. Voter interacts with the machine by using touch screen where he makes his choices and ballot is then printed from the BMD. This ballot is then ready to be cast. Voter can immediately check his ballot for any mistakes. If no mistakes were found, voter proceeds to optical scan and inserts marked ballot. Since unfilled symbols are filled by BMD, there is a small chance of the ballot not being counted. Using this separate marking device can also help disabled people to mark their ballots as they use additional accessibility software to ease the marking process.

Using optical scans allows for ballots to be counted faster and more precisely with less error. This means that there is no need for hiring that many people to take care of ballot counting and other activities required for managing the ballots.

Auditability

OMR systems are great combination of verifiable papers and electronics, because the voting process is more transparent. People understand it and trust it since it is similar in many ways to classical paper ballot method. This method is based on VVPAT so in case there is any suspicion of tampering with results, there is always an option to count physical paper ballots which can then be compared with results from OMR systems.

Security

Optical scans do not need internet connection and therefore they cannot be hacked remotely. The machine itself is built in such a way, that cast ballots are stored within a storage section of the machine. It is secured by multiple locks to which only election commission staff have required keys. Machine builders use security screws which should protect the machine from attackers opening them up. Tampering with optical scans would require tampering with physical ballots as well as with digital versions of them.

However, during DEF CON conference where part of it focused on voting machines and took place in Las Vegas, hackers revealed some vulnerabilities in DRE systems as well as optical scan machines. The vulnerabilities were mostly of physical matters. By using a screwdriver, hackers were able to get into the machine and replace its memory card. This would allow them to replace machine's operating system with their own and tampering with functionality of this scanning machine. They were also able to lock pick the ballot storage and steal all of the paper ballots. (*Hackers Highlight Vulnerabilities in Vote-Scanning Machines*, 2019)

4.2 I-voting

With massive adoption of internet in early 21st century people were inclined to find out ways how the internet could be integrated into the election process. During the last two decades many things have undergone digitalization, so why not the elections? The idea behind I-voting is that the voter could cast his votes using any device that has access to the internet such as smartphone, tablet, laptop or desktop computer. Meaning that voter could vote from anywhere in the world and would not have to be at a specific location at a

specific time in order to cast his vote. This idea is fascinating, however, many countries decided not to implement I-voting election systems as they think that it is not yet secure enough for handling voting. Some countries have already tried to run I-voting elections under some limited conditions. Countries that were the first to try I-voting were Estonia, Switzerland, France and the Philippines. (René Peralta, 2016)

4.2.1 Security risks

Estonia and DDoS attacks

Estonia is probably the most interesting one from the countries that have tried I-voting, as they have already developed significant infrastructure for digital democracy during the last several years. However, even though their infrastructure is highly developed there were cases when it was massively attacked by distributed denial-of-service attacks. (René Peralta, 2016) This resulted in a disruption of the use of the internet as well as shutting down websites of all government ministries, two major banks and several political parties. (Herzog, 2011) These attacks happened by the end of April 2007 after “Estonian government moved a memorial commemorating the Soviet liberation of the country from Nazis to a less prominent and visible location in Tallinn.” (Herzog, 2011) Estonia was then continually cyber attacked for almost 3 weeks. Primary Estonian officials immediately accused Russia of being behind these attacks, as it seemed like obvious reaction to the relocation of the memorial. However, technical experts from NATO and European commission were unable to link these DDoS attacks to Kremlin as there was no credible evidence of such actions. (Herzog, 2011)

Cyber-terrorists behind these attacks hacked many home desktop computers in countries like Egypt, Russia and the United States and by using swarming DDoS strategy they flooded Estonian digital infrastructure with enormous number of requests from these computers. Websites of banks and governments that usually have around 1000 visits a day could not handle the traffic of 2000 requests per second. (Herzog, 2011) Because of this, Estonia decided to preserve traditional paper ballot voting method alongside the I-voting possibility as well. (René Peralta, 2016)

Malware (malicious software) in voter's devices

Second main security concern is the potential malicious software that could infect voter's personal devices which they would use to cast their votes. This malware could then tamper with the vote while casting it in an undetected way, and no one would ever find out that the vote has been tampered with. (René Peralta, 2016) This malware could infect the voter's devices by visiting a website or by downloading an infected file. The attacker could replicate some of the most visited websites in particular region and target bigger percentage of voters, not only few individuals.

Vote manipulation

A third concern is related to possibility of vote selling, coercion and also family voting. These potential problems come with the fact, that I-voting takes place in an uncontrolled environment so there is no one except the voters themselves who could prevent these problems from happening. In case a voter would be forced to vote by someone, Estonian voting system allows the voters to vote multiple times during election days using the

voting app. By using the I-voting app they can vote as many times as they want but only the last vote counts as valid and their previous votes are invalidated and stored in logs for further analysis. They also have an option to go and vote traditional way in voting polls, but once they vote this way the cast vote is final, and they cannot revert it even if they voted again in the voting app. Some might say that people voting through the internet have more rights as they can vote multiple times if they change their mind and, in some way, it is true, but it is the way they chose to resolve potential coercion since people affected by it usually vote in an uncontrolled location and secrecy might not always be taken for granted. Therefore, they must have multiple ways at disposal for voters to cast their votes.

Even though there is some space to lure people into selling their votes, Estonia has not yet encountered any major vote sellos probably due to the fact that votes cast by voting app are not final and can be casted again. Buyer is therefore not guaranteed a vote for his desired candidate, unless he makes a deal with a voter to cast a paper ballot vote in person.

4.2.2 Functionality of Estonian I-voting system

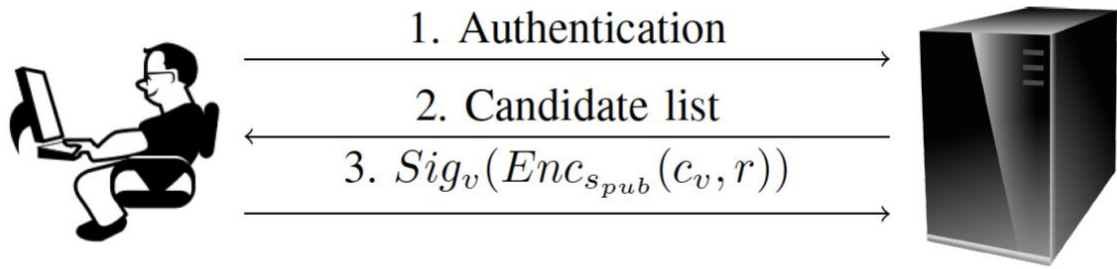
Since Estonia is so digitally focused country, they can do almost anything from anywhere when it comes to public services. This includes things like filling annual taxes, signing documents, daily banking, buying parking tickets etc. They can even create paperless EU company and be able to run it from anywhere in the world with minimal hassle. This is possible due to Estonian digital identities which are incorporated in ID cards or special SIM cards thanks to which they can prove themselves online as well as provide legally binding RSA signatures online. (Rang, 2019) (Heiberg and Willemson, 2014)

Things needed for I-voting:

- **I-voting software** – this software is downloaded from official government site and then installed on the computer. This way they eliminate any risks involved in using browser to vote.
- **Digital ID card / Mobile-ID** – Estonians can either use digital ID cards or they can use their mobile phones with special SIM cards that have the ID certificates embedded into it. This is called Mobile-ID and it is available through Estonian phone contracts. It is just a matter of preference which way suits you better as the result of authentication is the same in the end.
- **2 PIN codes** – they also have 2 separate PIN codes for authenticating and giving a digital signature which are needed throughout the voting process

Voting process is then quite simple. First step is to authenticate their identity. After opening the voting app, they enter their phone number and personal ID code. Both computer and mobile phone then display verification codes that have to match. After confirming a match, voter enters PIN 1 into mobile to authenticate and that brings them into the computer app where their personal and voting details are displayed. Next step is to choose a candidate and continue. Verification codes are displayed again on both devices and should match. After accepting matching codes on mobile phone, voter is asked to enter PIN 2 to digitally sign the vote and finish the voting process. By finishing the last step, the vote is cast and should be successfully recorded by digital ballot box. (Rang, 2019)

In a more technical detail, the ballot casting scheme on conceptual level is quite simple and is very similar to double envelope posting just in digital manner. The inner envelope represents the vote itself where voter's choice is stored. Central voting system generates an RSA key pair public and private key, where public key S_{pub} is available for the application and later used to sign the vote. By using an ID card or mobile ID voter v is able to authenticate himself and receive candidate list. Then makes a choice c_v which is represented by a number of a candidate and is signed and encrypted with the public key S_{pub} . RSA-OAEP encryption is used along with a random seed (salt) r to encrypt the vote. All these parts make an inner envelope which is computed as $b_{anon} = Enc_{S_{pub}}(c_v, r)$. The inner envelope is then wrapped by outer envelope which is created by signing the vote with voter's digital ID. This completes the ballot to its final form $b = Sig_v(b_{anon})$ which is then sent to a server. (Heiberg and Willemson, 2014) Below we can see a basic scheme of this voting protocol (see picture 6).



Picture 7: The basic Estonian voting protocol (Heiberg and Willemson, 2014)

These digital ballots are then stored in a digital ballot box in their encrypted form. When time comes to tally all the votes and voting period is over, voters signatures are dropped, this ensures vote secrecy, and the ballot itself which contains the candidate is decrypted with private key of a server. All of these ballots are then stored within a hardware security module.

4.2.3 Auditability

Until 2013, the Estonian voting system did not have any sort of auditability of votes cast using I-voting app. Verifiable I-voting system was then proposed and formed due to several attacks that happened in 2011 which tried to revoke the results of the whole electronic voting as well as the fact that voters had no way to verify if their votes were cast and recorded properly with their desired candidates (Heiberg and Willemson, 2014)

After finishing the voting process, they can download a separate application on their devices from Electoral Commission to verify their vote. They can simply open the app, scan a QR code which is created and displayed in the computer voting application after casting a ballot and within few seconds they can see if the vote was submitted with a correct candidate or not. Also, I-voting ends before polling day itself, which is another precaution in case of some problems with the I-voting system or potential attacks to manipulate elections. In case of such events which could threaten integrity of elections Estonian Electoral Commission would stop the I-voting completely, nullify all cast I-votes, and everyone would have to go and vote in person at voting polls. (Rang, 2019)

4.3 Paper Ballot method

As previously mentioned, paper ballots are used in the vast majority of the countries in the world including Czech Republic. However, this method is also not perfect in every aspect of security measures and certain things could be solved by taking advantage of implementing blockchain technology into the voting process. Paper ballots are used in current election system in Czech Republic so this voting method could potentially be replaced in the future if government was to decide to modernize current voting system.

The biggest concern regarding paper ballot voting is corruption and fraudulent behaviour resulting in mass manipulation with the outcome of the election. Cases of corruption and election manipulation are quite common in developing countries that are more authoritarian.

4.3.1 Belarus presidential election in 2020

One of the most recent cases of such actions was the presidential election in Belarus, which took place on the Sunday 9th of August 2020. Alexander Lukashenko, who has been in power for almost 27 years, had to face an opposition opponent Sviatlana Tsikhanouskaya. According to experts, this was the fight of his life because he clearly did not have the supporting majority of the population. He simply could not allow himself to be defeated by a woman. His victory would in this case have to be resounding and unequivocal with at least 80 per cent of the votes in his favour as any other outcome with less dominant result would be humiliating for him, said Valerij Karbalevich author of the book on Lukashenko in an interview with Euronews. (*Who is Belarus' long-time leader Alexander Lukashenko?*, 2020)

Final results of elections were declared by The Central Electoral Commission which stated that Mr Lukashenko won by acquiring astounding 80.1% of the vote and Ms Tsikhanouskaya only 10.12%. Ms Tsikhanouskaya claims that on voting polls where ballots were properly counted, she had support of 60% to 70% of the voters. Next day on Monday she went to register complaint about the results of election and was held for 7 hours. Right after that she was forced into exile. (*BBC News*, 2020) The results were also condemned by the EU and the US. Massive protests started across the whole country, mainly in Minsk, right after polls were closed and results were published. People were demanding Lukashenka's resignation and transfer of power. (*BBC News*, 2020) This resulted in a brutal crackdown against the protesters which was launched by Lukashenko himself when he ordered police to clear the streets after continuous rallies against him. (*Belarus: EU imposes sanctions as Lukashenko orders police to clear the streets*, 2020) (Welle (www.dw.com), 2020)

According to platforms Golos (Voice), Zubr (Bison) and the Honest People initiative the official CEC's data are not credible because they do not correlate with official data received from the polling stations. Golos platform had 1 261 127 registered users and 1 047 933 users with verified phone numbers who used this platform to post their votes to reflect on their actual choices. Their final data show that 95,58% (1 001 569) of verified users voted for Sviatlana Tsikhanouskaya. Detailed report can be seen on their official website. (*Golos platform presents the final report on the presidential election*, 2020)

Shenanigans

During this election people witnessed lots of suspicious behaviour from electoral commissions. Some saw and recorded members of electoral commission climbing down the ladder from upper floor of school building with bags full of votes. All of this was happening under supervision of a police officer who held the ladder from down below. (*Five Takeaways From The Belarusian Presidential Vote*, no date) This definitely does not give a credible impression of true and honest election. Others who were chosen to volunteer in polling stations were also witnesses of dishonest manners happening right in front of their eyes during preliminary as well as main voting on Sunday.

According to these volunteers, it seemed like the preliminary was dominated mostly by Lukashenko in majority of the polls. This might be due to electoral commission counting non-existent ballots of people who did not come to vote because the volunteer recorded around 20 people who actually voted that day and in final results there were 70 votes in total. A friend of his also told him that someone voted for his deceased father at another voting poll which was explained to him as a random mistake. At the end of the voting day volunteers were given a paper with results which they had to sign. Some were hesitant because they clearly knew what was happening, but they were also scared and had no idea what would happen to them if they refused to sign the results. Especially students were worried so that they would not have to face any problems during their studies on paid universities. In the end they signed, albeit reluctantly. (*How Belarus voted Minsk election commission members explain the shenanigans and intimidation used to stage Alexander Lukashenko's 'landslide victory'*, no date)

What volunteers would also agree on is that during main voting on Sunday majority of the voters wore white ribbons which is a symbol of opposition. These people also had a specific way of folding their ballots that identified them as votes against Lukashenko. In one case, around 80% of the votes were folded like accordions. Chairwoman and her secretary would then start sealing votes without even counting how many people came to vote that day first. When someone step up and said something against these practices, they just silently got rid of him. At this poll however, final results were more than 500 votes for the president and just 100 votes for Tsikhanouskaya. Obviously, they pulled these results out of thin air and even forced everyone to sign them, said "Konstantine" one of the volunteers. At the end of the day most of the cast ballots were in favour of Tsikhanouskaya at the time of counting them, but once CEC published final results, they were simply modified in their desired favour. (*How Belarus voted Minsk election commission members explain the shenanigans and intimidation used to stage Alexander Lukashenko's 'landslide victory'*, no date)

To sum it up, this election was clearly manipulated from the very beginning by the government under the command of President Lukashenko. Results of this election give the impression that none of the candidates had even a slightest chance of winning because they clearly had all the electoral commissions under control as well as Central Election Commission. This type of manipulation can happen in any other country which uses paper ballot method; however, it is unlikely to happen in true democratic countries.

4.3.2 Problems that could be solved by implementing blockchain

- Corruption and manipulation (huge problem in less democratic countries)

- Ballot stuffing by election officials (more votes than people who voted)
 - Publishing falsified results of election
 - Absentee ballots
 - Votes cast in the names of deceased people
 - Double voting problem
- Transparent ballot counting and tallying
- Faster and more accurate ballot counting (results within minutes/hours)
- Better accessibility for disabled and foreign people
- Less people required to run an election
- Lower costs of elections in the long-term

5 Czech election system

In the Czech Republic, the two most significant elections are Presidential and Chamber of Deputies of the Parliament elections (hereinafter referred to as “Parliament election”). Parliament election is defined in act “č. 247/1995 Sb. - Zákon o volbách do Parlamentu České republiky a o změně a doplnění některých dalších zákonů, aktuální znění 11.02.2021 (verze 31)”.

In this section the author will try to describe main phases of parliament election as well as all necessary parts of it to better understand all processes behind current voting system. Understanding this will help author to properly design a system which will try to preserve most of currently established processes and methods. To do so, the author will analyse current system in 3 phases in which most elections can be divided.

- Election preparation phase
- The conduct of the elections
- Ballot counting, verifying, and publishing election results

5.1 Election preparation phase

5.1.1 General provisions

Parliament elections take place once in every 4 years. That is the period for how long political parties are elected. The Parliament of the Czech Republic is elected on the basis of universal, equal and direct suffrage by secret ballot method. Current system uses the principles of proportional representation. Proportional system works best for this type of election, it will be preserved in the final design with no changes needed. Parliament election is announced by the president no later than 90 days before their holding. Election last 2 days and take place on Friday and Saturday so people have as much free time to go and vote. In foreign countries it also lasts 2 days but take place on Thursday and Friday. Eligible voter is a citizen of Czech Republic who has reached the age of at least 18 years during the second day of voting. (info@aion.cz, no date a)

Constituencies

For voting and counting ballots there are permanent constituencies created in each municipality. Constituencies are created to cover around 1000 voters. Some might be bigger or smaller dependent on a location where they are situated. Constituencies cannot be changed from announcement of election until results are published by State election commission. (info@aion.cz, no date b) In Czech republic there are 6253 municipalities in 77 state districts which are part of 14 state regions. (*Česká republika - Obce, okresy a kraje kdekoliv*, no date) During last Parliament election there were 14865 constituencies available. (*Celkové výsledky hlasování | volby.cz*, no date)

Permanent list of voters

All eligible voters are recorded either in a permanent list or in a special list. Permanent list of voters is maintained by a municipality or a city district in which voters have permanent residence. Each voter can be recorded in only 1 of the lists at a time. Voter can verify if he is present on a permanent list at municipal office and can require filling in additional information or making necessary changes. 2 days before the first election day at 16:00, municipal offices close the permanent lists and hand over an extracts of these lists to constituency election commissions. (info@aion.cz, no date b) These extracts let's call them "voter lists" are then used in constituencies to verify and record eligible voters.

Special list of voters

Special list of voters contains voters who:

- don't have permanent residence in a particular district
- are in a hospital or institute of similar manner
- are in a custody or in jail
- vote using voter's card
- live abroad and are citizens of Czech Republic

These lists are closed and finalized at least 20 days before election. Extracts from the lists are sent to district electoral commissions.

Voter card

Voters can request a voter card at their municipal office if they know that they will not be able to vote in their constituency during election days. This card allows them to vote in any constituency in Czech Republic. Once they acquire this card they are moved from a permanent list to a special list of voters. (info@aion.cz, no date a)

State electoral commission

State electoral commission is a permanent electoral authority which organises the whole election. Its job is overseeing and securing the actions necessary to run everything smoothly. In the end, this authority publishes final election results in Collection of Laws. These results are as well handed over to the relevant chamber of the Parliament. (info@aion.cz, no date a)

District election commission

This is the commission which sits in the polling place. Its role is to authenticate voters, oversee voting, count ballots, and prepare a final report about the course and results of an election at their constituency. (info@aion.cz, no date a)

Polling station

According to paragraph 17 of act "č. 247/1995 Sb." each polling station must be equipped with a non-transparent ballot box, a portable ballot box, voting ballot papers, blank envelopes bearing an official stamps and extracts from permanent and special lists of

voters. There are also several voting booths and writing accessories so that voters can mark their choices and seal an envelope with a ballot in private. (info@aion.cz, no date a)

Ballots and Preferable votes

Each political party has their own ballot which contains a list of their candidates who will represent the party in the Parliament in the end. This list is sorted by the most important party candidates to less important ones. However due to preferable votes mechanics, voters have an option to give 4 preferable votes to the party candidates, who they like the most by marking them on the ballot in case they would like to see them represent the party in the Parliament. Voters may not like the leaders who are at the top of the list, therefore this mechanics allows them to have impact on the structure of the political party as well. ('Preferenční hlasy a preferenční hlasování', 2018)

5.2 The conduct of the election

Right before an opening of the vote, a chairman of the election committee must check if the polling station is fully equipped according to paragraph 17 as well as check if the ballot box and the portable ballot box are empty. The chairman then seals the ballot boxes in front of the rest of the election commission. Once everything is ready and set, the chairman declares the vote open. Voting usually looks like:

1. Voter enters the polling station and goes to an election commission member
2. Election committee member asks the voter for an ID to check if he is present in the list of voters and eligible to vote. Election member also verifies if the voter shows his personal ID or not.
3. Voter is then given a ballot (if he already does not have one) and an envelope with an official stamp.
4. Voter then goes to a voting booth and chooses appropriate political party ballot in which he may or may not mark his preferable votes and then places it into the envelope and seals it.
5. Voter leaves the voting booth and puts the envelope into a ballot box which is situated in front of election commission.
6. Ballot is now cast, and voter leaves the polling station.

Special cases

1. There might be a case when a voter comes to a polling station and is not present in a voter list, but he proves with his ID that he is eligible to vote there. In this case election commission manually adds him to a voter list and allows him to vote. (info@aion.cz, no date a)
2. People with specific circumstances, primarily related to health, can ask municipal office or election commission to allow them to vote out of the controlled polling station. In this case the district election commission sends two election commission members with a portable ballot box, an official envelope, and voting ballots to a voter. Commission members then proceed in a way so that secret vote is preserved. (info@aion.cz, no date a)

3. Voter who votes using a voter card gives this card to an election commission, which attaches it to an extract from a special list of voters. Voter then normally proceeds with voting. (info@aion.cz, no date a)
4. People abroad vote on special constituencies created on Czech embassies and consulates.

After the first election day election commission seals the ballot boxes so no ballots can be inserted and taken out and secures all the documentation as well. At the start of the second election day election commission checks integrity of the seals on the boxes and removes them. Once the second election day ends, polling stations are closed, and the chairman of the district election commission declares the vote closed. (info@aion.cz, no date a)

5.3 Ballot counting, verifying and publishing election results

Once the voting has ended the chairman of election commission seals all non-used ballots and envelopes. After that he opens both ballot boxes and election commission take out all the ballots which are mixed immediately. Invalid ballots and envelopes are excluded, and the rest of the valid envelopes is counted and compared to the number of voters from extracts from permanent and special lists of voters to see how many people voted and how many envelopes they have. Ballots are then taken out of envelopes, divided by political parties, and counted. Counting is overseed by the chairman of the district election commission. Members of state election commission can also be present during counting. (info@aion.cz, no date a)

When the counting is done and results are available, the election commission prepares two identical reports about the course and the outcome of the vote. This report is signed by every member of the election commission. One report is handed over to Czech Statistical Office which then process it further if the report does not contain any mistakes. Czech Statistical Office then process the reports from all the constituencies and prepares detailed reports for each of 14 regions. Based on the results of each region, the Czech Statistical Office uses a formula to count total mandates that will be split between regions. Lastly, the State Electoral Commission prepares a final report from all the regional reports. The Final report consists of (info@aion.cz, no date a):

- Total number of people recorded in extracts from permanent and special lists of voters
- Total number of voters who were given official envelopes
- Total number of valid votes for each political party, political movements and coalitions separately and for each region separately
- Names of people sorted by political parties who were elected and candidates who became alternates

By publishing these results in the Collection of Laws, the Parliament election is finished.

6 Blockchain election system design proposal

This section will be dedicated to describing a blockchain election system design, which should give an insight on how such a digitalized election system could look like in the future. Firstly, the author will define criteria which needs to be met by the system. Then the most appropriate voting technology will be chosen followed by description of the proposed system itself and its key components. Used technologies appropriate for this proposal are then described in the following section.

6.1 Blockchain election system criteria and prerequisite

Criteria

The system needs to comply with all the international election standards discussed in chapter 3.2. Proposed system should meet these main criteria:

1. Preserve secrecy of the vote
2. Everyone has equal number of votes
3. System is tamper proved
4. System is transparent
5. System is accessible to everyone

Prerequisite

The Czech Republic currently is not in the same spot as previously discussed Estonia which has been able to develop significant digital infrastructure over the years, but more importantly, they managed to get digital governance adopted by Estonians. Vast majority of the people in the Czech Republic are still used to go and handle all the paperwork in person in different offices even though all ID cards from mid 2018 are embed with microchips that allow for digital use of ID cards. (*Občanský průkaz s čipem | EOP | Informační web elektronické identity*, no date)

ID card can be used mainly for online identification as well as digital signing of documents and accessing 3rd party applications. The problem is that the ID card is not ready by default to be comfortably used online by its owner and it requires some technical knowledge to set it up properly to use it online as well as buying additional hardware and certificates to use the full potential of the ID card. Owner must have 3 PIN codes where 2 of them are used to manage identification function and another 3 separate PIN codes if he wants to use the other features. Remembering 6 PIN codes is not practical at all and it might be one of the reasons why it not widely used yet. (*Jak začít využívat identifikační funkci občanského průkazu | EOP | Informační web elektronické identity*, no date), (Pe, no date)

Right now, most people do not know that these new electronic ID cards exist, or how to use them. Adopting eGovernance and daily use of these new ID cards might take several

years, which is also the timeframe in which a blockchain election system of a similar manner to the one proposed in this thesis might potentially be implemented.

6.2 Choosing the right voting technology

Previously, we have discussed few different voting technologies that are used around the world. OMR machine would be great choice in general, but they do what they need to do thanks to the way they are designed. It also does not solve the potential problem with corruption, as the election staff might simply add additional ballots into the machine. There is not much room for further digitalization of this technology. More appealing voting technologies for our system would be DRE machines and i-Voting. Both options offer great opportunities for digitalizing voting.

i-Voting

I-Voting would be, without a doubt, the most comfortable way of voting. However, to create a fully digitalized election, i-Voting is not an appropriate solution because we cannot assume that older generations would know how these technologies work. I-Voting would make the system not easily accessible to everyone. We can see that in Estonia, it is also an option for certain segment of people who are comfortable with it, but paper ballots are still preserved. In their case, the paper ballots solve the problem of coercion as well as offer easily accessible way of voting to the people in a familiar way. I-Voting could however be a nice addition to an already digitalized system which is proposed in this thesis.

DRE machines

For fully digitalized election system a DRE machine would be the best choice. The voting process using this technology is quite similar to the traditional paper way, because it would be used instead of a ballot box in a controlled area at a polling station. Because casting votes would still take place in polling stations, the people have their privacy and cannot be coerced. Having these machines would also help disabled people to make the election process as much accessible to them as possible.

DRE machine would be equipped with an ID card reader used for authenticating into the system. A great addition would also be a VVPAT machine which can offer some form of reassurance to the voter that the ballot was cast as intended. The paper created by VVPAT would however not be considered as a valid paper ballot rather it should serve as a receipt to check the correctness of the cast vote.

Machines would have access to the internet by being connected to a local router. The network in a polling station can be setup in a way, that all the communication that would come from the outside are forbidden and only communication out would be allowed from the machines. Voting machines would not even be visible on the network, so no one could reach the machines through the internet. This would prevent any potential DDoS attacks. Tampering with already cast votes on blockchain is impossible even if an attacker hacked the machine.

To interact with the blockchain the voting machine needs to be connected to the internet the whole time. We cannot get rid of internet connection completely because we wouldn't be able to cast the votes on a blockchain immediately. Not being connected to the internet and recording the votes locally in the machine would also be quite risky. The machine

could break down and all the cast votes stored inside would be inevitably lost. There could be some backup solution which would allow the system to be functional offline, but for the purpose of this proposal let's assume, that the network is secured properly.

Each polling station should be equipped with at least 2 voting machines. Dependent on the size of a constituency numbers of needed machines could vary. These machines would also be present in constituencies abroad for Czech citizens living or being outside of the country during election.

DRE machine would also

6.3 Proposed system

For this system author presumes that everyone has an electronic ID card and remembers a PIN code to access its identification feature. ID card is one of the key components in this system because everyone must have one by the law and it offers the easiest and most accessible way of proving your identity online. Utilizing it in a blockchain voting would only make it more useful.

6.3.1 Digital wallet

Digital wallet is another key component of this system. It is used to interact with blockchain by being able to send transactions from one digital wallet to another. One wallet can interact with many different blockchains, so its use is not limited for this system only as there might be many other use cases for blockchain digital wallet in the future. Digital wallet can store many cryptocurrencies and tokens which can be used for different purposes. In this case, it would store the votes itself in a form of tokens.

Each voter would be given access to his digital wallet at a polling station after physically proving his identity to an election commission. Election commission would then generate a digital wallet by using NIZKP protocol. This wallet could then be accessed by authenticating with the ID card in a voting machine. This approach is inspired by proposed design by Hjálmarsson and Hreiðarsson. (Hjálmarsson *et al.*, 2018)

6.3.2 Ballots

As mentioned above, ballots would have a form of tokens. To have as much control over the tokens as possible, we must create custom tokens where we can configure desired supply of the tokens available as well as other parameters.

For simplicity, the system could only use 1 type of a token, but then we could only determine how many votes each candidate acquired, because all the tokens are exactly the same. We would be able to pick parties and candidates who acquired most of the votes, but we could not tell from which constituency, municipality, district, and region the votes were cast. Because of this, mandates which are counted based on results of each region could not be counted as well.

To be able to determine from where the votes were cast, we would have to create different tokens ideally for each constituency, because it is the lowest level at which the votes are counted. This would provide us with the ability to prepare detailed results reports by the end of election. During the creation of the tokens, information about constituency, municipality, district, and region under which this token will be counted can be added.

This information would be stored as a metadata in the token itself, thanks to which results of each region could be counted.

These tokens would then be provided to a voter after successful authentication into a voting machine.

6.3.3 Basic system logic

The basic idea behind the logic of this blockchain voting system is quite simple. Voter comes to a polling station and shows his ID card to an election commission based on which he is given access to a digital wallet which is loaded with tokens as soon as he authenticates into a voting machine. Political parties and each individual party candidate have their own unique digital wallets. Casting a vote happens by sending a token as a transaction from a digital wallet of a voter to a digital wallet of a desired political party and digital wallets of preferable candidates based on the choices voter makes in the voting machine. When election ends, final number of tokens from each constituency is already available on digital wallets of political parties and candidates. All of this can be seen on public blockchain ledger. At this stage Czech Statistical Office would have to further process these numbers by extracting them from blockchain and preparing results reports.

In the picture below we can see random transactions that happened in a block n. 12386586 on Ethereum blockchain. We can see to which block the transactions belong as well as digital wallets between which each of the transactions happened. All these transactions are irreversibly written on blockchain.

Transactions

For Block 12386586

A total of 247 transactions found

Txn Hash	Method	Block	Age	From	To	Value	Txn Fee
0x391e079557fe3b96d...	Transfer	12386586	35 secs ago	0x98e998c1da13ec699e...	0xa7be50cf885ebba9eff...	0.00506519 Ether	0.000756
0xbb473146e72e005af4...	Transfer	12386586	35 secs ago	0xd840ebc9b90e3af757...	0xa376b5c0c81952c3ff2...	0.05 Ether	0.000756
0xf65023f4c9aaf388446...	Multicall	12386586	35 secs ago	0x4558ef17b852c4c19b...	Uniswap V3: Router	0 Ether	0.00469008
0x97d22f2ad1fb17523b0...	Transfer	12386586	35 secs ago	0xf8cef5c660af25d9d60...	0x20e648dae2e0ea6d4...	0.02877 Ether	0.000756
0xb8235019330b751665...	Transfer	12386586	35 secs ago	0x04c5d153a1b09633e2...	Tether: USDT Stablecoin	0 Ether	0.00165992
0x445f7c507693c986a7...	Transfer	12386586	35 secs ago	0xe152d5f653c9ecf4108...	Tether: USDT Stablecoin	0 Ether	0.00227509
0xd4758a4c0bb40f1b95...	Transfer	12386586	35 secs ago	0xd713f49b161cf41d7ed...	Tether: USDT Stablecoin	0 Ether	0.00227552

Picture 8: Ethereum transactions in a random blockchain block (source: Author)

6.4 Used technologies

6.4.1 Public Blockchain

The system could be developed either on a public or a private blockchain. The main difference between the two is that private blockchain is governed by a single entity which makes the blockchain centralized. This governing entity can structure the blockchain in a way that not everything is publicly available to everyone. Only parties which are given permissions to read, write or audit the blockchain can access it. (*Public vs private blockchain: Who wins?* | *public blockchain* |, 2020) If we were to deal with some form of a

corruption from the state level, private blockchain governed by the state could make it impossible to see what was happening during election and if the election was honest or not.

On the other hand, public blockchain is an open network which can be used by anyone and no single entity has power over it. Everything is available to everyone which makes it more transparent and gives less space for tampering with the election. Public blockchain is therefore ideal choice for building transparent elections as no transactions can be hidden there.

6.4.2 Ethereum Blockchain

Since we would not likely be implementing this system now, but rather several years in the future, one of the most appropriate blockchain to build this system on would be Ethereum blockchain. It is currently the most used blockchain for developing decentralized applications.

The reason why it would not be suitable solution now are high gas fees which users have to pay for making transactions on the network. Fees on Ethereum network these days can go up to even 20\$ for a single transaction which is not something voters would be willing to pay just to cast a vote. If we were to count how much it would cost based on the voter turnout from 2017 when last parliament election was held, this would make 101 892 660\$ just for using Ethereum network. (*Celkové výsledky hlasování | volby.cz*, no date) It is unlikely that even the state would be willing to endow this amount of money.

The fees are so high is because the Ethereum network can now process only around 15 transactions per second to this day and the demand behind it is much higher, which makes the network overloaded. However, Ethereum is looking forward to a huge upgrade to Ethereum 2.0 during year 2021. This upgrade will improve the network's scalability which will allow it to process around 100 000 transactions per second. (*ETH 2.0: Confessions of a Sharding Skeptic*, 2020) Such a speed of the network would be more than enough for our proposed system and it would also make the transaction gas fees negligible.

6.4.3 Tatum

Developing decentralized applications directly on any blockchain can be time consuming and expensive because most of the applications which use smart contracts are developed in Solidity programming language. Solidity is popular among blockchain developers, but compared to other programming languages, its community is way smaller roughly around 200 000 developers compared to for example JavaScript which has at least 9.7 million worldwide. (*The most popular programming languages used in blockchain development*, 2019)

The application could therefore be developed on a Tatum platform. Tatum is a free REST API blockchain infrastructure platform. It requires no blockchain expertise by offering simple and unified REST API. This platform takes care of many things for its users such as maintaining blockchain infrastructure and node configuration. Developers can then focus on the development of their decentralized apps in an easy way rather than dealing with problems regarding blockchain itself. It offers support of over 20 different blockchain protocols including Ethereum. (*Welcome to Tatum* 🙌, 2021)

Things like creating digital wallets, custom tokens or sending transactions are then just a few lines of easily understandable code.

6.4.4 Authentication system in a voting machine

Authentication system would allow only eligible voters to access the voting system with their ID card. Voter needs to know their PIN code which is between 4 to 14 numbers long to be able to authenticate into the voting system and be able to cast their votes. In case voter would enter 3 incorrect PINs his ID card would get blocked. He would then have to go to a city hall to have it unblocked or he could unblock it himself if he knew the second PIN code which is used to unblock your ID card.

List of eligible voters would be acquired from national registers of citizens the same way permanent and special lists of voters are prepared now. Anyone who not eligible to vote would not be able to get into the voting system.

Security

We can assume that a thief could steal some ID cards from people. ID cards on their own would be useless to him because he would also have to know the PIN codes used for authenticating. And even if he knew the PIN codes, he would still need to present the physical ID to an election commission to confirm his identity based on which he would be created a digital wallet which is essential for casting votes.

6.4.5 Smart Contract

By definition “A smart contract is a self-executing contract with the terms of the agreement between buyer and seller being directly written into lines of code. The code controls the execution, and transactions are trackable and irreversible.” (Frankenfield, no date) Proposed application would have to contain a smart contract which would take care of certain things.

This smart contract would take care of sending the votes. There would be a function called “vote” which would contain digital wallets of political parties and candidates to which tokens need to be sent after voter made his choices in a voting machine.

Smart contract would also take care of loading a voter’s digital wallet with appropriate tokens based on voter’s permanent residence. This would happen as an on-demand transaction immediately after the voter would successfully authenticate into the voting system. For parliament election every voter would be loaded with 5 tokens in total, 1 token for a political party and 4 tokens for preferable candidates. Thanks to having different tokens for each constituency, voters would not be limited to vote only in an assigned constituency, but they could vote from any other polling station in the Czech Republic and their votes would still be properly counted under the right constituency.

To ensure that an authority which would take care of creating this smart contract is not planning to misuse its power to tamper with the election, there should also be at least 3 other independent authorities which would examine the final smart contract before the election to check if there is no questionable code which could tamper the election. These independent authorities would also oversee the election itself for suspicious behaviour.

6.4.6 Non-interactive zero knowledge proof protocol

NIZKP or non-interactive zero knowledge proof protocol is variant of zero knowledge proof. This tool is used between two parties and it allows them to prove that they know certain piece of data without revealing any specific information other than the value itself without the need of any interaction between each other. It uses a reference string which is common and shared between both parties based on which they can achieve Zero Knowledge proof. (Villalobos, Altamirano and Chandra, no date)

Non-interactive zero knowledge proof protocol would be used during the creation of digital wallets of the voters at a polling station by election commission. This protocol would allow to create a digital wallet for a voter without linking his identity to it. By using an ID card for authentication voter could then access his created digital wallet and cast his votes secretly without anyone being able to link a specific wallet to his identity. By using this technology, the system would comply with the 1st criterion defined in chapter 6.1.

6.5 Conduct of the proposed election system

Conduct

- 1) Voter comes to a polling station and presents his ID card to an election commission
- 2) Election commission member checks the identity of a voter and creates him a digital wallet by using NIZKP protocol.
- 3) Voter comes to a voting machine and uses his ID and his PIN code to successfully authenticate into the system.
- 4) Once authenticated, the smart contract automatically sends appropriate tokens based on a residence of a voter as an on-demand transaction to a voter's digital wallet.
- 5) Voter interacts with the application in the machine and selects desired political party
 - Voter can still decide to return back and not cast any votes
- 6) Voter is shown a list of party candidates in case he wants to select preferable votes as well.
 - Preferable votes are not mandatory
- 7) Voter selects 1 to 4 candidates of his choice and confirms to cast the votes.
- 8) Tokens are sent to the chosen parties and candidates as transactions on Ethereum blockchain.
- 9) Election is done and voter leaves the polling station.

Since everything can be seen live on the blockchain, some people might argue that seeing results live and not keeping the results away from public until the end of election could affect other voters who are yet about to vote. This is true and it makes a good point, however, only qualified authorities would know which wallets belong to which parties and candidates. Therefore, results are theoretically available on the blockchain immediately,

but no one except the qualified authorities would be able to gather real results until the end of election. Someone could make a brutal analysis of the movement on the blockchain but still would be only able to make assumptions unless he knew specific information about the wallets.

Counting votes

Once election days are over results are already available on digital wallets of each political party and candidates. Data from the blockchain would then have to be extracted from the blockchain by Czech Statistical Office to prepare results report based on which State Election Commission could publish final and official results of the parliament election in the Collection of Laws.

6.6 Other possible use cases

In Czech Republic there are many other different elections such as presidential, senate or European Parliament elections. Proposed design requires more developed digital infrastructure as well as better adoption of technologies such as electronic ID cards and eGovernment. Proposed system could definitely be modified to suit other nation-wide elections as well.

However, for any small local elections for example municipal council elections, it would probably be better to stick with the traditional way of voting, until blockchain technologies evolve even further and it is easier to adapt it to different situations. Already functional voting system such as Helios, which is ideal for smaller elections, might also be a great choice for this purpose.

7 Conclusion

7.1 Evaluation of the proposed design

In chapter 6.1, the author has defined system criteria for the proposed system design.

The first and most important criterium is the preservation of the secrecy of the vote. This was achieved by using Non-Interactive Zero Knowledge Proof protocol during digital wallet creation. Thanks to that, digital wallets cannot be linked to the voter's identity. This allows them to cast their voting using blockchain election system in complete secrecy, which is one of the principals of voting in democratic elections.

Equal number of votes was achieved by defining the number of tokens in a smart contract, which would be sent to a voter only after successful authentication into the voting system. By defining it in a smart contract, there is no possible way of changing it once the smart contract is created. It would however have to be inspected by independent authorities to verify the validity of the contract and its compliance with the laws. Everyone would have the same number of available votes with no possibility of tampering with them. This is a huge improvement over the traditional paper ballot system currently used in Czech Republic, because when using paper ballots, we cannot truly address any of the manipulation problems mentioned in chapter 4.3.2.

By recording every vote directly on the blockchain, it makes it impossible to tamper with already cast votes. This cannot be said about paper ballots cast into a ballot box. Incorporating ID cards into the system also helps to minimize potential tampering and unauthorized access from election commission or anyone else.

The last two criteria were related to election transparency and accessibility. Having the system run on a public blockchain makes it transparent to everyone and less prone to manipulation, because everything is trackable right on the network. The design complies with the accessibility criteria as well because the proposed design does not require you to buy any additional hardware, neither it does function in a non-understandable way. By using DRE machines instead of ballot boxes in the polling stations, the voting process stays nearly the same which makes it easier to get adopted by people.

Proposed system design therefore complies with all the system criteria that has been defined as well as with all international election standards.

7.2 Achieved goals

In the beginning, the author has set several goals for this thesis:

- Gather enough resources in the first theoretical part of the thesis which can be used for detailed description of the proposed system and its implementation.
 - Give an overview of a blockchain technology and voting systems using this technology
 - Define elections and election standards

- Provide analysis of different voting technologies used in the world
- Describe current Czech Parliament election system
- Propose blockchain election system design for Czech Parliament elections

Theoretical and analysis parts of the thesis were covered in chapters 2, 3, 4, 5. Information provided, gathered, and analysed in these chapters provided the author with enough knowledge to design Czech Parliament blockchain election system described in chapter 6. All the goals set in the beginning of this thesis can be therefore considered as accomplished.

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